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HW#16 Stat 50

Section 4.11

(5) $\mu = 15$, $\sigma = 5/\sqrt{60}$

(a) $P(\bar{x} < 14) = P\left(\frac{\bar{x} - 15}{5/\sqrt{60}} < \frac{-1}{5/\sqrt{60}}\right)$

$= P\left(Z < \frac{-\sqrt{60}}{5}\right) = P(Z < -1.5492)$

$= 0.0606$

(b) $P\left(\frac{\bar{x} - \mu}{5/\sqrt{60}} < Z < \frac{\bar{x} - \mu}{5/\sqrt{60}}\right) = 0.7000$

$\frac{\bar{x} - \mu}{5/\sqrt{60}} = 0.5250$

$\bar{x} = \left[0.5250\left(\frac{5}{\sqrt{60}}\right)\right] + 15$

$\boxed{\bar{x} = 15.34}$

(c) $P(\bar{x} < 14) = 0.01$

$P\left(\frac{\bar{x} - 15}{5/\sqrt{n}} < \frac{14 - 15}{5/\sqrt{n}}\right) = 0.01$

$P\left(Z < \frac{-1}{5/\sqrt{n}}\right) = 0.01$

(2)

$$-2.325 = -\frac{1}{5/\sqrt{n}}$$

$$-2.325 = -\frac{\sqrt{n}}{5}$$

$$\sqrt{n} = 11.625$$

$$\boxed{n = 136}$$

(7)

(a) $\mu = 4.0$

$\sigma = 2$

$$P(S_{50} < 180) = P\left(Z < \frac{180 - 200}{14.14}\right)$$

$$= P(Z < -1.414)$$

$$\boxed{= 0.0793}$$

(b) $$P\left(S_{50} < \frac{x - 4(\mu)}{\sqrt{n} * 4}\right) = 0.3000$$

$$\frac{x - 200}{\sqrt{200}} = -0.525$$

$$x = (-0.525 * \sqrt{200}) + 200$$

$$\boxed{x = 193}$$

(18) $\mu_1 = 0.5$ $\sigma_1 = 0.4$

$\mu_2 = 0.6$ $\sigma_2 = 0.5$

$n = 100$

(a) $P(\bar{x}_1 > 55) = 1 - P\left(Z \leq \frac{55 - 0.5(100)}{\sqrt{(.4)^2 \times 100}}\right)$

$= 1 - P(Z \leq 1.25)$

$= 1 - 0.8944 = 0.1056$

$P(\bar{x} > 55) = 0.1056$

(b) $P(\bar{x}_2 \leq 55) = P\left(Z \leq \frac{55 - 60}{5}\right)$

$= P(Z \leq -1) = 0.1587$

(c) $\mu = 50 + 60 = 110$

$\sigma = \sqrt{4^2 + 5^2} = \sqrt{16 + 25} = \sqrt{41}$

$P(\bar{x} > 115) = 1 - P(\bar{x} \leq 115) = 1 - P\left(Z \leq \frac{115 - 110}{\sqrt{41}}\right)$

$= 1 - P(Z \leq 1.086) = 1 - 0.8611$

$= 0.1389$

(2) $\mu = (50 - 60) = -10$

$\sigma = \sqrt{41}$

$P(\bar{x} - (-10) / \sqrt{41}) \Rightarrow P(\bar{x} > 0) = P(Z > 2.44)$

$P(Z > 2.44) = 0.0073$

Section 5.1

(1)

(a) 95 %.

$Z_{\alpha/2} \Rightarrow$

Area $= 1 - 0.05 = 0.95$

$\alpha = 0.05$

$Z_{0.025} = 1.96$

(b) 98 %. Area $= 1 - 0.02 = 0.98$

$\alpha = 0.02$

$Z_{0.01} = 2.325$

(c) 99 %.

Area $= 1 - 0.01 = 0.99$

$\alpha = 0.005$

$Z_{0.005} = 2.575$

(4) 80%.

Area = 1 − 0.20 = 0.80

$\alpha = 0.20$

$\boxed{Z_{0.80} = 1.28}$

(5) (0.88, 1.02) ⇒ 95% C.I. , n = 114

(2) $\bar{x} = 0.95 \Rightarrow \frac{1.02 + 0.88}{2} = 0.95$

$\boxed{\bar{x} = 0.95}$

(5) $s = \left[\frac{1.02 - 0.88}{2(1.96)}\right] = 0.38132422 3$

$\Rightarrow \boxed{s = 0.3813}$

(7)

n = 80

(a) $\bar{x} = 1217$

$s = 52$

$P(Z < z) = 0.95 \Rightarrow z = 1.96$

$z \frac{s}{\sqrt{80}} \Rightarrow 1217 \pm 1.96 \left(\frac{52}{\sqrt{80}}\right)$

∴ 95% interval at (1205.6, 1228.4)

(b) 98%.

$P(Z < z) = 0.98$

$\boxed{Z = 2.33}$

98% C.I. $\Rightarrow 1217 \pm 2.33\left(\frac{50}{\sqrt{80}}\right)$

$\Rightarrow (1203.5, 1230.5)$

(c) (1208, 1226)

$1217 \pm 9 \Rightarrow 9 = x\left(\frac{50}{\sqrt{80}}\right)$

$x = 1.548047061$

~~95.45% CI~~

$\boxed{87.88\%}$